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1. Problem Statement

The Task API currently allows any client to update a task via `PUT /api/tasks/{id}` without any check against the version of the task the client last read. If two clients read the same task and both submit updates, the second write silently overwrites the first ("lost update"), with no indication to either client that a conflicting change occurred.

2. Proposed Change

Introduce optimistic concurrency control on `Task` updates. Each task will carry a version number, returned to clients in every response. A client updating a task must submit the version it last observed. If that version no longer matches the task's current version in the database, the update is rejected and the client is told to re-fetch the task and retry.

This does not change how tasks are created, listed, filtered, or deleted — only the update path gains a concurrency check.

3. Scenario

1. User A opens Task #42 ("Prepare report"), currently at version 1.
2. User B, on a different device, also opens Task #42 at version 1.
3. User A changes the status to `IN_PROGRESS` and saves. The save succeeds; the task is now at version 2.
4. User B, unaware of User A's change, edits the due date and saves — still submitting version 1, since that's what User B's screen last showed.
5. The API detects that the submitted version (1) does not match the task's current version (2), and rejects User B's update with `409 Conflict` instead of silently discarding User A's status change.
6. User B's client re-fetches the task (now showing User A's change at version 2), re-applies the due-date edit, and saves successfully — advancing the task to version 3.

4. Functional Requirements

- **FR-01:** Every `Task` shall carry a version number that increments by exactly 1 on each successful update.
- **FR-02:** Every API response that returns a task (create, get, list, update) shall include its current version.
- **FR-03:** `PUT /api/tasks/{id}` shall require the client to submit the version it last observed, alongside the other task fields.
- **FR-04:** If the submitted version does not match the task's current version at the time of the update, the API shall reject the request with `409 Conflict` and shall not apply any of the submitted changes.
- **FR-05:** If the submitted version matches, the update shall proceed and the version shall advance by 1.
- **FR-06:** The conflict response shall clearly state that the update was rejected due to a version mismatch, consistent with the project's existing error response format.
- **FR-07:** Existing tasks (created before this change) shall be assigned an initial version automatically; no manual data migration shall be required for existing rows.

5. Acceptance Criteria

- A `PUT` request submitted with the current version succeeds and the returned task's version has incremented by 1.
- A `PUT` request submitted with a stale (outdated) version is rejected with `409 Conflict`, and the task's data in the database is unchanged.
- All task-returning endpoints (`POST`, `GET` single, `GET` list, `PUT`) include the version field in their response.
- Two genuinely concurrent update attempts against the same task result in exactly one success and one `409 Conflict` — never two silent successes.
- All existing Task and Category test scenarios that do not involve concurrent updates continue to pass unmodified.

6. Scope

In Scope:

- Adding a version field to the `Task` entity and its API responses.
- Requiring and validating the expected version on `Task` updates.
- Rejecting stale updates with `409 Conflict`.

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